

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	15 July 2026
Team ID	
Project Name	UCab Premium – Cab Booking Application
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration as Rider, Driver, or Admin Registration through Email & Password Email confirmation on sign-up
FR-2	User Authentication	Login with JWT-based token issuance Role-based access control (Rider/Driver/Admin) Secure password hashing
FR-3	Booking Management	Create a new ride booking View live booking status Cancel a booking with reason
FR-4	Driver Assignment	Assign nearest available driver to a booking Driver accepts/rejects assigned ride Update ride status (Assigned → Accepted → On the Way → Completed)
FR-5	Fare & Payment	Calculate estimated fare before booking Generate final fare on trip completion Generate PDF receipt (PDFKit)
FR-6	Ratings & Reviews	Rider rates driver after trip completion Driver views aggregate rating
FR-7	Admin Analytics	View ride, revenue, and user-growth charts (Recharts) Manage users and drivers View all bookings across the platform

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, role-specific dashboards for Rider, Driver, and Admin; responsive UI built with React and Vite for fast load times.
NFR-2	Security	JWT-based authentication, role-based access control, hashed passwords, and validated API inputs across all 35 REST endpoints.
NFR-3	Reliability	Consistent booking state machine (Pending → Driver Assigned → Accepted → On the Way → Completed/Cancelled) prevents inconsistent ride states.
NFR-4	Performance	MongoDB indexing on frequently queried fields (user ID, booking status) for fast lookups; lightweight REST responses.
NFR-5	Availability	Node.js/Express backend designed to run continuously with MongoDB Community Server; error handling and graceful failure responses.
NFR-6	Scalability	MVC-structured Express backend and modular MongoDB collections allow the system to move from local hosting to cloud/MongoDB Atlas as user load grows.

